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Twist buckling and the foldable cylinder: an exercise in origami

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Abstract

Basic mechanisms for the buckling of a thin cylindrical shell under torsional loading are reviewed from a post-buckling perspective. Deflections are considered so far into the large-deflection range that the shell is allowed to fold to a flat two-dimensional form, in a mechanism reminiscent of a deployable structure. Critical and initial post-buckling effects are explored through concepts of energy minimization and hidden symmetries. For comparisons with the final large-deflection folded shape, a truss element program is employed. It is shown that, as buckling develops, the mode shape must change to accommodate both the symmetry-breaking aspects of the predominately inwards deflection, and the rotation of peak and valley lines of the buckle pattern necessary to accommodate the geometry of the final folded shape.
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1. Introduction

As with buckling under axial compression, the buckling of the thin circular cylindrical shell under torsion is recognized as one of the fundamental problems of elastic stability [1–3]. But the two contrasting loading situations show quite different characteristics of load/deflection response as deformation continues into the large-deflection range. While the former has initially a highly unstable response, followed by restabilization [4] as it settles into a localized form of the well-known *Yoshimura* or *diamond* pattern [5,6] the latter forms similar but oblique shapes, which are capable of folding entirely flat in the axial direction without significant in-plane stretching. Fig. 1 shows the development of these folds in a paper specimen twisted between two inner plastic mandrels [7]. Following an initial buckling stage that involves both bending and membrane (in-plane) stretching, a pattern more like a mechanism from origami develops, after which the system can be folded completely flat as shown. Such folding is denied to the axially loaded Yoshimura pattern, where significant stretching occurs at the restabilization stage [8]. Folding without stretching is, of course, of considerable interest to the field of deployable structures [9–11].

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Fig. 1. Foldable cylinder based on twist buckling of a paper roll.

This paper explores the relationship between initial buckling, first-order post-buckling, and the final folding mechanism, in the twisted cylindrical shell. For the initial buckling, a simplified modeshape is introduced in terms of parameters measuring the wavelength and obliqueness of a circumferentially periodic wave. The resulting critical load is then minimized with respect to these parameters to give a close approximation to the preferred modeshape. For the first-order post-buckling, this solution is appended by a second axi-symmetric mode, as suggested by an initial symmetry-breaking combination found for the Yoshimura pattern [12]. The combined solution then compares favourably with that found experimentally [3]. For the folding mechanism, the large-deflection response is compared with that of a simple triangulated truss-like structure [13].

The sequence demonstrates that the circumferential wave number is decided at the point of instability via a linear eigenvalue problem, and remains unchanged throughout the folding process. However, other characteristics of the initial modeshape are obliged to alter as the system moves to accommodate the fixed geometry of the final folded shape. In particular, the obliqueness of the chosen pattern must change. This process involves the crest and valley lines of the deflection pattern starting parallel to one another but rotating by different amounts on the surface of the shell, until they meet at the ends of the buckle length to create the fold lines of the mechanism. This rotation of fold lines can be clearly seen in the twisted paper specimens.

2. Experiments

2.1. Folded paper

A simple experiment is shown in Fig. 1, based on a sheet of paper twisted between two plastic mandrels.

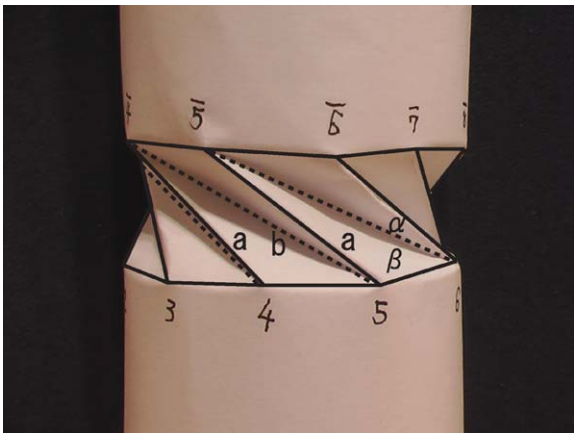


Fig. 2. Fold lines and angles of the folding mechanism.

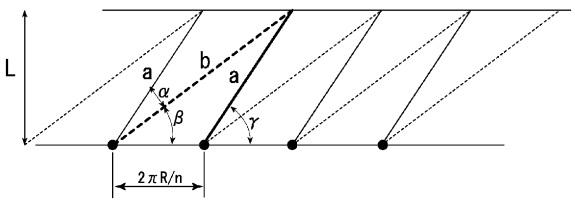


Fig. 3. Truss geometry.

Fig. 2 picks out the folds in this mechanism as either outwards (solid) or inwards (dashed) and the significant angles α and β and γ are shown schematically in Fig. 3. Here α is the apex angle for a typical triangle of the mechanism, β represents the obliqueness of the inwards fold, and $\gamma = \alpha + \beta$ the obliqueness of the outwards fold, as shown. The simple geometry of Fig. 3 shows

$$\sin\gamma = \sin(\alpha+\beta) = \frac{L}{a}, \quad \frac{a}{\sin\beta} = \frac{2\pi R}{n\sin\alpha},$$

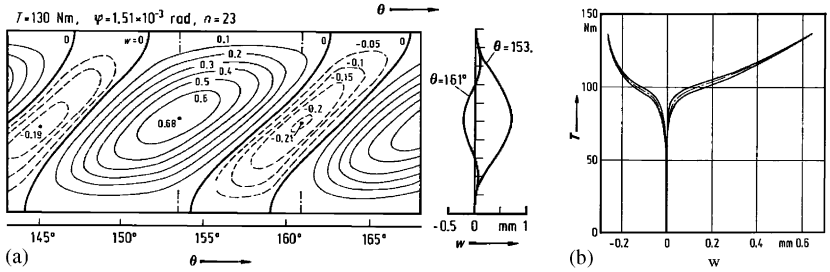


Fig. 4. Yamaki's shortest cylinder (length $L=22.9$ mm): (a) contour of deformation, (b) relationship between torque and $w_{\max}$. We appreciate that both Springer-Verlag and Prof. Junji Tani of Tohoku University, who have permitted to use the Yamaki copyright for us, willingly.

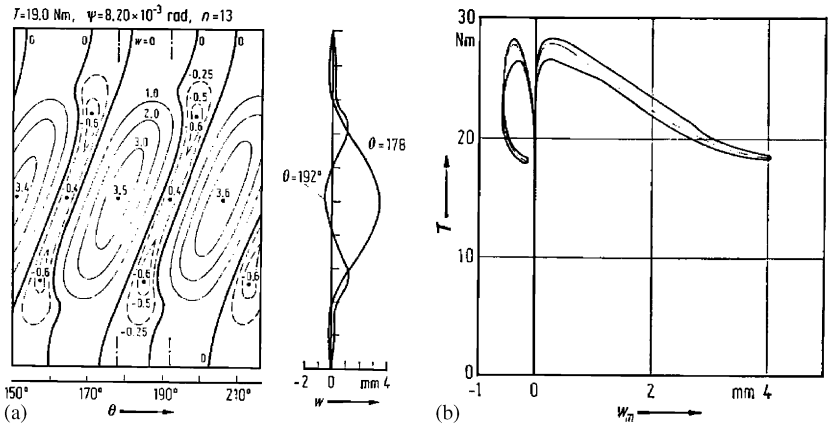


Fig. 5. Yamaki's longest cylinder ($L=160.8$ mm): (a) contour of deformation, (b) relationship between torque and $w_{\max}$. We appreciate that both Springer-Verlag and Prof. Junji Tani of Tohoku University, who have permitted to use the Yamaki copyright for us, willingly.

where L is the axial length of the buckle and R is the radius. This can be rearranged for the constraint equation

$$\sin\gamma = \sin(\alpha+\beta) = \frac{nL}{2\pi R}\frac{\sin\alpha}{\sin\beta}. \tag{1}$$

If the apex is to fold to a point on the circumference without stretching along the fold lines, this equation is further supplemented by the fixed definition of α. Euclidean geometry indicates that α equals one-half the angle subtended at the centre by the same chord length, and if there are n full waves around the circumference, then [14],

$$\alpha = \frac{\pi}{n}. \tag{2}$$

Thus if n is given, α, β and γ are all uniquely defined by Eqs. (1) and (2).

2.2. Experiments of Yamaki

Yamaki [3] published a series of experiments on thin polyester cylindrical shells of radius $R=100$ mm, thickness $t=0.247$ mm, Young's modulus $E=5.55$ GPa and Poisson's ratio $\nu=0.3$, of different lengths under torsion. He pursued deflection well into the post-buckling range, and mapped (in one case at more than one stage) the deflected shape as contours. Figs. 4 and 5 show various torque/deflection plots and deflection maps for the shortest and longest cylinders, respectively. Shell dimensions, together with those for intermediate length cylinders, are given in

Table 1
Experimental results from Yamaki [2,3] ($R = 100\,\text{mm}$, $t = 0.247\,\text{mm}$)

Yamaki shell no.	20-5	50-4	100-4	200-4	500-4	500-4	1000-4
L (mm)	22.9	35.9	51.0	71.2	113.8	113.8	160.8
$Z = \sqrt{1-\nu^2}L^2/Rt$	20.2	49.8	100.3	195.8	500.2	500.2	998.6
n	23	21	19	17	15	15	13
γ (deg.)	44	56	59	67	73	69	70
β (deg.)	30	46	52	62	69	65	65
$\alpha = \gamma - \beta$ (deg.)	14	10	7	5	4	4	5

Table 1. We see that the short cylinder buckles in a stable-symmetric manner [15], while the long shell shows unstable-symmetric buckling with a degree of axial localization into the centre [16]. In each case, deflection is predominately inwards.

The contours of Figs. 4 and 5, etc. [3] show the pattern of deformation at a stage somewhere between initial buckling and final folding. Later, we model the initial buckling as periodic folds, with obliquely orientated crest and valley lines that run parallel to each other. It is clear from attempting to add to the experimental plots of Figs. 4 and 5 the best hand-drawn approximations to these lines, that by the stage shown crest and valley lines are no longer parallel; the system is already beginning to accommodate the folding mechanism shown in Fig. 3. In Table 1, we have designated the best guess for the orientation of the outwards and inwards (crest and valley) lines taken from the plots of Yamaki as γ and β, respectively, in accordance with Fig. 3. The Rayleigh–Ritz analysis which follows is able to accommodate only equal rotations of crest and valley, and hence $\beta = \gamma$ ($\alpha = 0$) always in the analytical comparisons that follow. Two sets of values are given for shell 500-4 because the deformation pattern is mapped twice, the second set being at a slightly later stage in the loading process.

3. Shell formulation

The total potential energy for a thin cylindrical shell of length L, radius R and thickness t of a material of Young's modulus E and Poisson's ratio ν, loaded in torsion by shear stress τ acting circumferentially along the boundaries $x = 0, L$, can be written as [8],

$$\begin{aligned} V = {} & \frac{Et^3}{24(1-\nu^2)} \\ & \times \int_0^L \int_0^{2\pi R} \{(\nabla^2 w)^2 + 2(1-\nu) \\ & \times [(w_{xy})^2 - w_{xx}w_{yy}]\}\,\mathrm{d}y\,\mathrm{d}x \\ & + \frac{Et}{2}\int_0^L \int_0^{2\pi R} \{(\nabla^2\phi)^2 + 2(1+\nu)[(\phi_{xy})^2 \\ & - \phi_{xx}\phi_{yy}]\}\,\mathrm{d}y\,\mathrm{d}x \\ & - \tau t \int_0^L \int_0^{2\pi R} w_x w_y\,\mathrm{d}y\,\mathrm{d}x \\ & - Et\int_0^L \int_0^{2\pi R} \phi[\nabla^4\phi + \rho w_{xx} - (w_{xy})^2 \\ & + w_{xx}w_{yy}]\,\mathrm{d}y\,\mathrm{d}x, \end{aligned} \tag{3}$$

where $w(x, y)$ is the radial deflection, $\phi(x, y)$ is a stress function, ∇^2 and ∇^4 are, respectively, the two-dimensional harmonic and bi-harmonic operators, and the subscripts denote partial differentiation with respect to the axial coordinate x and circumferential coordinate y. Here the first term is the bending energy, the second is the membrane energy, the third is the work done by the externally applied torque $T = 2\pi R^2\tau t$, and the fourth is an extra energy term resulting from a constraint condition that links the stress function ϕ with the change in Gaussian curvature of the surface described by w. By minimizing with respect to arbitrary shapes for w and ϕ, the corresponding von Kármán–Donnell equations

$$\begin{aligned} \kappa^2\nabla^4 w + \mathscr{T} w_{xy} - \rho\phi_{xx} = {} & w_{xx}\phi_{yy} + w_{yy}\phi_{xx} \\ & - 2w_{xy}\phi_{xy}, \end{aligned} \tag{4}$$

$$\nabla^4\phi + \rho w_{xx} = (w_{xy})^2 - w_{xx}w_{yy} \tag{5}$$

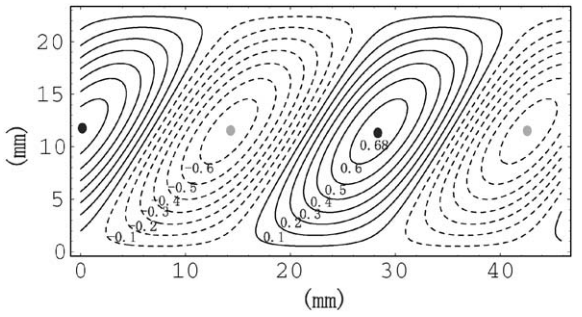


Fig. 6. Contours of the simply-supported buckling mode, plotted for $R = 100\,\text{mm}$, $L = 22.9\,\text{mm}$, $n = 22$ and $\lambda = 3.07$.

are obtained, where $\rho = 1/R$ is the curvature, $\kappa^2 = Et^2/12(1-\nu^2)$, and the bifurcation parameter $\mathscr{T} = \tau/E$. The first is an equilibrium equation and the second is the compatibility equation tied to the constraint condition.

4. Rayleigh–Ritz analyses for critical buckling

In this section, we present Rayleigh–Ritz analyses that seem to provide good estimates of critical wave number n, the critical torque T^{C}, and the obliqueness measure λ as defined below. As with all such analyses, the quality of the result depends on the quality of the approximation used for the buckled modeshape. Here two alternative formulations are given based on the two limiting flexural boundary conditions of simple and clamped supports. Both approximations compare reasonably well with the experiments of Yamaki [3], with the clamped conditions showing slightly superior results.

4.1. Simply supported end conditions

If the ends of the buckled length are simply supported in bending, an appropriate modeshape is,

$$w(x,y) = A\,\sin\left(\frac{\pi x}{L}\right)\sin\left(\frac{\lambda x}{L}-\frac{ny}{R}\right), \tag{6}$$

$$\phi(x,y) = B\,\sin\left(\frac{\pi x}{L}\right)\sin\left(\frac{\lambda x}{L}-\frac{ny}{R}\right), \tag{7}$$

where A and B are unknown amplitudes. Contours of this modeshape are shown in Fig. 6. Substituting these into the energy function V of (3) and minimizing with respect to A and B leads to two equilibrium equations which can be solved as an eigenvalue problem for the critical torque T^{C}. This critical value can then be minimized by successive inspection with respect to both n and λ, to give a theoretical value for T^{C} available for comparison with the experiments of Yamaki [3].

4.2. Clamped end conditions

If the ends of the buckled length are clamped against bending, an appropriate modeshape is

$$w(x,y) = A\,\sin^2\left(\frac{\pi x}{L}\right)\sin\left(\frac{\lambda x}{L}-\frac{ny}{R}\right), \tag{8}$$

$$\phi(x,y) = B\,\sin^2\left(\frac{\pi x}{L}\right)\sin\left(\frac{\lambda x}{L}-\frac{ny}{R}\right). \tag{9}$$

These can be handled in precisely the same manner as for simple supports.

4.3. Comparisons with experiments of Yamaki

Comparisons between the results of the two alternative Rayleigh–Ritz formulations are given in Table 2. Here, the experimental values for n and γ are repeated from Table 1 together with estimates of critical torque T^{C}. These are compared with analytical predictions from the two alternative Rayleigh–Ritz formulations, produced using the algebraic manipulation software Mathematica.

5. Rayleigh–Ritz solutions for initial post-buckling

Cylinder buckling is characterized by the need to break hidden symmetries. The initial buckling modes, as seen in Fig. 6 and represented by Eqs. (7) and (9), are flip invariant [17]—reversing amplitudes A and B gives the same energy—while the deflection pattern as seen in Fig. 2 clearly is not. The initial post-buckling would thus be expected to demonstrate a strong symmetry-breaking character, in the sense as seen in Figs. 2, 4 and 5 that inwards deflection is considerably more pronounced than outwards. One successful way of admitting such (hidden) symmetry-breaking is with the addition of an axial mode with half the axial wavelength of the initial buckling mode

Table 2
Comparisons between linear eigenvalue results and the experiments of Yamaki [2,3]

Yamaki shell no.		20-5	50-4	100-4	200-4	500-4	1000-4
Experiment	T^C (N m)	99	64	51	41	33	28
	n	23	21	19	17	15	13
	γ (deg.)	44	56	59	67	73 or 69	70
Simple supports as given by Eqs. (6) and (7)	T^C (N m)	93.2	62.2	48.1	38.8	29.4	24.3
	n	22	20	18	16	13	11
	λ	3.07	3.59	3.88	4.07	4.10	4.05
	γ (deg.)	59	63	67	70	75	77
Clamped supports as given by Eqs. (8) and (9)	T^C (N m)	115.7	71.3	53.6	42.6	32.0	26.3
	n	23	21	19	17	14	12
	λ	3.50	4.11	4.45	4.70	4.83	4.87
	γ (deg.)	56	61	65	69	73	76

[18], as follows:

$$w(x,y) = A_1 \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{\lambda x}{L} - \frac{ny}{R}\right) + A_2 \sin^2\left(\frac{\pi x}{L}\right),$$

$$\phi(x,y) = B_1 \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{\lambda x}{L} - \frac{ny}{R}\right) + B_2 \sin^2\left(\frac{\pi x}{L}\right). \quad (10)$$

The combination of buckling modes A_i, B_i generates non-linear post-buckling equations, $\partial V/\partial A_i = \partial V/\partial B_i = 0$, which can be solved using algebraic manipulation software to obtain the post-buckling solution. At the same time, minimization of V is carried out with respect to the orientation variable λ, to allow for rotation of peak and valley lines as seen in the paper samples. A typical modal combination is shown in the contour plot of Fig. 7. In contrast to Fig. 6, the contour plot shows many of the characteristics of the experimental contours of Yamaki [3]; having broken the hidden symmetry, the deflected shape has the significant property that inwards deflection (the elliptical shape) is considerably more pronounced than outwards (the dumb-bell shape). Unlike in the paper samples, however, the pattern maintains peak and valley lines as parallel, rather than allowing them to rotate by different amounts as observed. For this reason, we refer to the equilibrium paths obtained as those of *initial post-buckling*.

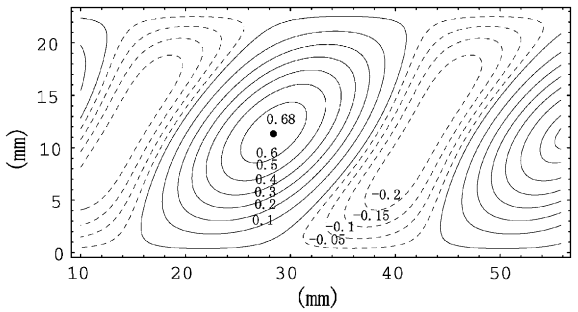


Fig. 7. Contours of initial post-buckling, plotted for $R = 100$ mm, $L = 22.9$ mm, $n = 22$ and $\lambda = 3.07$.

5.1. Comparisons with experiments of Yamaki

Fig. 8 compares the initial post-buckling curves of torque T plotted against maximum inwards and outwards displacements, with experimental results for Yamaki's [3] longest and shortest cylinders. Simply supported boundary conditions are assumed for each theoretical plot. Fig. 8(b) for the long cylinder shows a reasonable comparison over the range of the experiment, but for the shortest cylinder of Fig. 8(a) the comparison is less good. Here, the experiments show a stable form of buckling with torque rising as the displacement increases, while the theoretical result has the torque initially falling and then starting to rise.

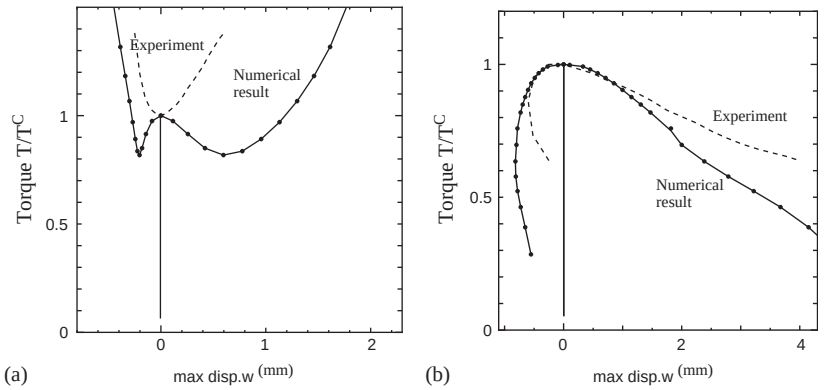


Fig. 8. Comparison between results of initial post-buckling and experiments: (a) shortest cylinder ($Z \approx 20$), (b) longest cylinder ($Z \approx 1000$).

Note, however, that after this initial mismatch, generalized stiffnesses, as measured by the slope of the post-buckling curve, match the experiment well in both the positive and negative displacement sense.

We attribute the difference mainly to a mismatch of boundary conditions at the ends of the buckled region. The assumed shapes of Eq. (10) match neither simple nor clamped supports at the ends, and seemingly allow too much freedom to the system, leading to a reduced post-buckling stiffness. The significantly better result for the long cylinder indicates that the effect reduces with length. Evidence of localization in the buckle pattern for the long cylinder would suggest that the boundary conditions at the ends of the buckled region have reduced significantly as the shell gets longer [16]. Moreover, there is clearly a limit to the length that can accommodate the geometry of Fig. 3. This again is likely to encourage localization to a more suitable buckle length.

Fig. 9 shows the relationship between the displacements $w_{\max}$ and $w_{\min}$, and the orientation angle γ, for the shortest and longest cylinders. We see that in each case the value of γ initially rises slightly, before decreasing as the post-buckling starts to accommodate the geometry of the folding mechanism as seen in Figs. 1 and 2. The effect is more pronounced for the short than the long cylinder, reflecting the fact that the short cylinders are obliged to make much greater changes of orientation between initial buckling and the final folded shape. This can be seen clearly in the results of the truss analysis which follows.

6. Truss analysis

It is instructive to compare the buckling behaviour as seen in Figs. 1 and 2 with the response of a triangulated truss structure comprising just tension/compression members and pin joints [13]. Although their actual structural characteristics differ in a number of significant ways, the geometries of the two systems show a number of common characteristics over the large deformations of the folding process. Most importantly, if we assume the fold lines of Fig. 3 to be inextensional, the observed triangulated structure is closely analogous to the truss model behaving as a pure mechanism, with a zero energy input requirement in the form of work done by torque. For the shell, apart from local effects at the ends of the buckled region, the same triangulated form would involve bending in the absence of stretching and thus would demand only a small energy input. This zero or low energy requirement contrasts with classical shell buckling under axial compression, which provides effective resistance against load both before buckling, and in the advanced post-buckling range in the known presence of a plateau or *Maxwell* load [19]. Similar behaviour over the initial post-buckling range is found both experimentally and theoretically for short shells only, as described above. Thus, early in the buckling process, we expect the shell to behave in the manner of the buckling and initial post-buckling analyses presented earlier, while in the later stages as it approaches the deployable shape indicated in

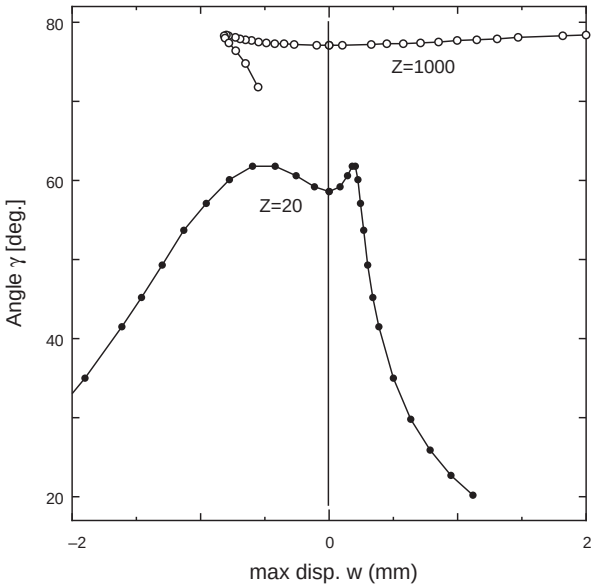


Fig. 9. Relationship between γ and w for Yamaki's shortest ($Z \approx 20$) and longest ($Z \approx 1000$) cylinders.

Fig. 1(c), it should behave more like a truss model mechanism. The process of adaptation between the two stages is clearly more severe for short than for long cylinders.

Eqs. (1) and (2) hold for both shell and truss models, and in particular define the geometry required for the truss to behave as or nearly as deployable mechanism. They are valid for cylinders of all lengths, yet it is apparent from the comparisons seen below that there is a significant switch in behaviour somewhere in the range $200 < Z < 500$ ($0.71 < L/R < 1.28$). Table 3 shows comparisons between the experimental wave numbers and orientations of Table 1, the linear eigenvalue predictions of Table 2, and the truss mechanism suggested by the special geometry of Eqs. (1) and (2). It is noticeable that the orientations measured both by the experimental and the linearized predictions of γ are reasonably close to the mechanism values for long shells, but for the shorter shells they are considerably different. This again suggests that the boundary conditions at the ends of the buckled region carry considerably more significance for the short than for the long shells. On the other hand, the longer shells are clearly more able to tolerate large applied twists without much change from the form of initial buckling. This becomes clear from comparing values of α in Table 3. At the stage seen experimentally, the long cylinders have peak and valley lines that remain close to being parallel, as demonstrated by the fact that the experimental values of α are considerably smaller than those seen in the folding mechanism.

The same change in characteristics between long and short cylinders can be seen in the following plots. Fig. 10 shows, for a shell of medium length, two comparative triangular patterns, one based on the simply supported initial buckling mode of Table 3, and the other on the truss mechanism. The increase in orientation angle γ required to produce a mechanism is clearly seen. On the other hand, Figs. 11(a)–(f) give torque to end-shortening plots for various truss models of all of Yamaki's [3] shells, relating to all four situations of Table 3. Whereas the "final buckling" mechanism shows that the full end-shortening can be achieved with very little torque for short and medium length models, this is not the case for the longer models of the lower two plots. Now a significant positive torque, followed by a reversal to a negative value, is required to collapse the section completely. This represents significant stretching and/or compression in the fold lines of the equivalent patterns to those of Fig. 1. This stretching effect becomes so intense by $Z \approx 1000$, that the geometry chosen by the experimental

Table 3
Comparisons between experiments, initial buckling analysis, and folding truss mechanism

	Specium no.	20-5	50-4	100-4	200-4	500-4	1000-4
Experiment	n	23	21	19	17	15	13
	γ	44	56	59	67	73 or 69	70
	β	30	46	52	62	69 or 65	65
	$\alpha=\gamma-\beta$	14	10	7	5	4	5
Simple supports	n	22	20	18	16	13	11
Initial buckling	γ	59	63	67	70	75	77
Clamped supports	n	23	21	19	17	14	12
Initial buckling	γ	56	61	65	69	73	76
Truss mechanism as given by Eqs. (1) and (2)	n	23	21	19	17	14	12
	γ	24	30	35	42	56	72
	β	16	21	26	32	43	57
	$\alpha=\gamma-\beta$	8	9	9	10	13	15

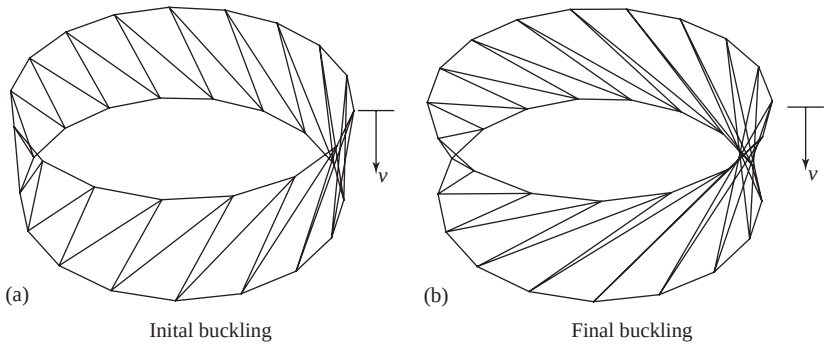


Fig. 10. A truss model for the Yamaki shell with $Z \approx 200$: (a) initial buckling, (b) final buckling.

configuration as seen in Fig. 5, if maintained over the whole collapse mechanism, seemingly would do less work over the full course of folding than the geometry chosen purely by the most suitable final form. Thus the final buckling mode is likely to prefer localization to a shorter buckle length, rather than remaining with the geometry dictated by the truss mechanism of Eqs. (1) and (2).

7. Concluding remarks

The circumferential shear introduced at the pre-buckling stage suggests that membrane compression is maximized at some oblique angle to the axial direction, and hence valley and peak lines in the buckle pattern take oblique orientations. It is interesting to compare the final folded form with the shape of the well-known Yoshimura or diamond buckle pattern related to the final deflected shape of a cylindrical shell under axial compression, as seen in Fig. 12, where the valley lines are perpendicular to the axial direction. The two patterns show many similarities, but also have significant differences. The Yoshimura shape is effectively locked into its unstretched state; under axial compression it represents a minimum on the load deflection curve and will continue to take extra load, albeit at a reduced stiffness compared with the pre-buckling value, after the minimum has been reached. The same is apparently true for short cylindrical shells under torsion. To force such systems completely flat would require a considerable amount of extra membrane stretching. The underlying reserve of strength associated with the minimum load value has recently been linked to the so-called *Maxwell load*, more commonly associated with the concept of phase change

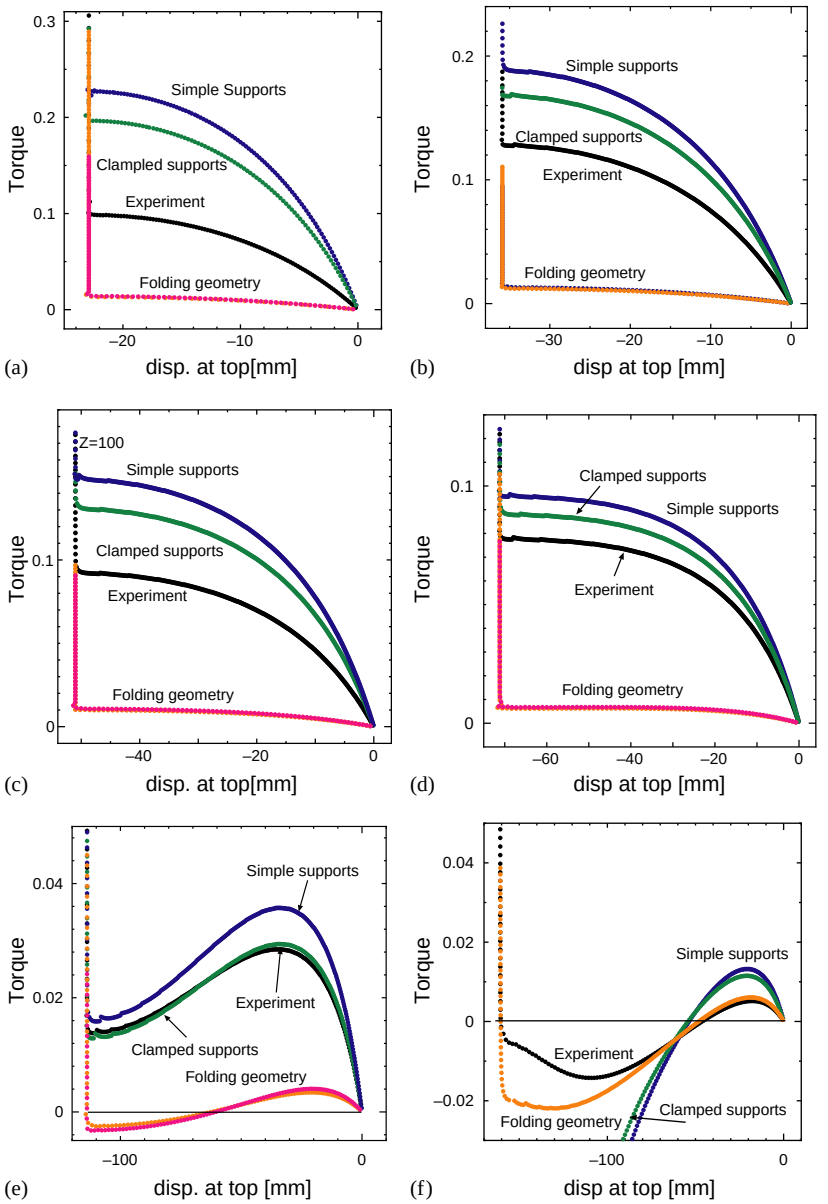


Fig. 11. Relationship between torque and vertical displacement v for different truss geometries: (a) $Z = 20$, (b) $Z = 50$, (c) $Z = 100$, (d) $Z = 200$, (e) $Z = 500$, and (f) $Z = 1000$.

in thermodynamical systems [19]. On the other hand, the twisted mechanism continues to fold to the flat state with little in the way of extra stretching. This accommodation is entirely a result of the oblique orientation. It is interesting to observe that some non-symmetric lay-ups in composite shells can also give oblique buckle patterns [20]. This suggests that there is a possibility of such complete folding taking place under axial compression, a circumstance ruled out in the isotropic shell by reasons of symmetry [21].

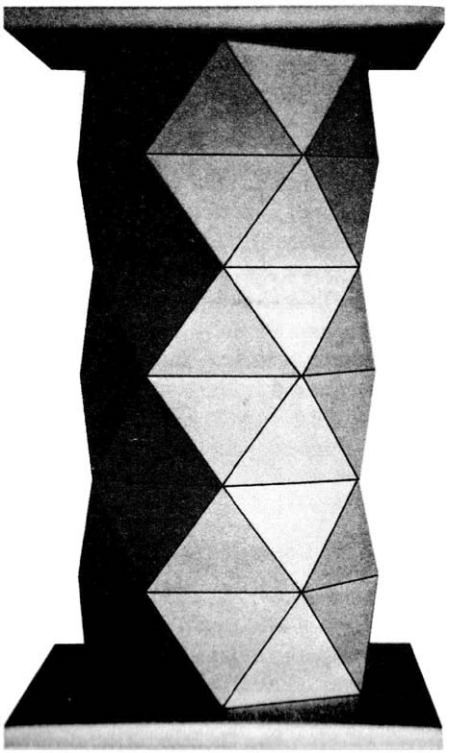


Fig. 12. A Yohsimura buckle pattern, folded from a flat sheet of card.

Acknowledgements and dedication

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